

Article

Potential of Carbon Offsets to Promote the Management of Capercaillie Lekking Sites in Finnish Forests

Aapo Tikka ¹ , Arto Haara ², Janne Miettinen ³, Piia Ikonen ^{1,4} and Ari Pappinen ^{1,*}
¹ School of Forest Sciences, Faculty of Science, Forestry and Technology, University of Eastern Finland (UEF), P.O. Box 111, FI-80101 Joensuu, Finland; aapo.tikka@uef.fi (A.T.); piia.ikonen@uef.fi (P.I.)

² Bioeconomy and Environment, Natural Resources Institute Finland (Luke), Yliopistokatu 6B, FI-80100 Joensuu, Finland; arto.haara@luke.fi

³ Natural Resources, Natural Resources Institute Finland (Luke), Paavo Havaksen tie 3, FI-90570 Oulu, Finland; janne.miettinen@luke.fi

⁴ Finnish Wildlife Agency, Kiekkotie 4, FI-70200 Kuopio, Finland

* Correspondence: ari.pappinen@uef.fi; Tel.: +358-50-438-2527

Abstract: Competition between the economic and natural assets of forests is emphasised in capercaillie lekking sites, which are often located within production forests. In this study, we assessed the potential use of carbon offsets as income for the management of capercaillie lekking sites. We ran simulations and optimisations for two alternative forest management scenarios in ten forest holdings located in five different regions of Finland. The size of each forest holding was approximately 30 hectares, of which 5 hectares was included in a lekking site. The *basic scenario* followed forest management recommendations, and the *carbon sequestration scenario* aimed to enhance biodiversity maintenance in a way that enabled carbon offsets to be obtained for the lekking site. We found that the decrease in revenue from timber sales was generally so small that the income from carbon offsets provided an economically better choice. Furthermore, the cost-effectiveness of carbon sequestration did not seem to be restricted to a specific location. The approach we introduced can be utilised in future decision making in the forestry sector to promote the coordination of nature management and forestry. Moreover, carbon pools should be considered more comprehensively in future research.

Keywords: lek management; wildlife management; carbon sequestration



Citation: Tikka, A.; Haara, A.; Miettinen, J.; Ikonen, P.; Pappinen, A. Potential of Carbon Offsets to Promote the Management of Capercaillie Lekking Sites in Finnish Forests. *Forests* **2023**, *14*, 2145. <https://doi.org/10.3390/f14112145>

Received: 7 August 2023

Revised: 13 October 2023

Accepted: 25 October 2023

Published: 28 October 2023



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1. Introduction

Globally, biodiversity loss and climate change have become enormous challenges that will affect the future of humanity and the entire biosphere. However, the role of forests is partly contradictory. For example, forests can be utilised as a source of renewable resources to decrease the use of fossil-fuel-based materials but simultaneously play a vital role in the global carbon cycle as a carbon sink [1]. Consequently, the management of forests has a substantial influence on the climate. Moreover, forests provide habitats for a wide range of species. While some boreal species are favoured by current forestry practices, the size of many other populations has decreased, and some species are now endangered [2]. Since the network of protected areas is not sufficiently extensive, and suitable habitat patches for many species are small and isolated, our understanding of the biological interactions in production forests is vital for the maintenance of their ecological values. Thus, nature management activities should be carried out as part of forest management in all production forests, especially in those sites with a large number of species or endangered species with demanding habitat requirements.

Capercaillie (*Tetrao urogallus*) are an umbrella species in northern coniferous forests [3,4]. Thus, the management of capercaillie habitats also benefits many other species. From the viewpoint of wildlife management, lekking sites belong to most of the important habitats that capercaillie need during their annual cycle. Lekking sites also have a high

ecological value for the general conservation management. Usually, lekking sites are found in conifer-dominated mixed forests or in pine forests, both of which should exhibit a substantial variation in tree density [5,6]. Most importantly, capercaillie prefer mature forests and thinning forests as lekking sites [7], which must provide adequate forest cover and shelter [8,9] and are most likely to be found in mosaic-like forest structures [10]. The preferred horizontal visibility in lekking sites is 20–50 m [5] and lekking sites should have a stable forest structure [11]. Furthermore, areas with extensive home ranges and population-scale connectivity are high priority for conservation activities [12], highlighting the importance of cooperation in the management of lekking sites [13,14].

Traditionally, revenues for forest owners are accrued from timber sales, which are driven by the intensive management of production forests. However, these management practices usually do not promote biodiversity [15,16] and climate aims. Thus, the maintenance of habitats or maximisation of carbon sequestration [17] may have an influence on forest revenues. As boreal forest areas are heavily dominated by production forests [18,19], capercaillie lekking sites are often found outside of conservation areas, which makes the competition between economic and natural assets even more tangible. Consequently, the identification of opportunities to compensate for possible losses in net revenues is key to the promotion of lek management in private forests. In general, cost-effective ways to promote the production of ecosystem services can be found by changing the philosophy that underpins forestry practices [20], which can also benefit the habitats of grouse species [21]. In addition, the possibility of receiving payments for ecosystem services [22] could compensate for economic losses, provided these payments are available.

Although payment mechanisms for carbon sequestration are still under development [23,24], the extension of the carbon markets may make carbon offsets available to private forest owners. This would provide a new way to receive income from forests instead of timber sales provided that cuttings are decreased, which might allow for the cost-effective maintenance of the characteristics of forested habitats. Furthermore, studies have shown that forest owners have a willingness to receive carbon sequestration payments for their forests [24–26]. Indeed, their application in capercaillie lekking sites could lead to a win–win situation, which would be advantageous both for the forest owner and for achieving biodiversity and climate aims. Thus, carbon offsets could promote the multiple uses of forests, the purpose of which is to combine two or more objectives in the use of forests, including wood and non-wood products. Under what circumstances, then, could carbon offsets compensate for the economic losses caused by decreased timber sales?

In this study, our aim was to assess the potential application of carbon offsets in the management of capercaillie lekking sites (as an example of a forest biodiversity hotspot). Simultaneously, the objective was to introduce an approach to the multiple uses of forests that combines biodiversity, climate, and economic objectives for future forestry management practices.

2. Materials and Methods

In this study, two alternative forest management scenarios were assessed from the viewpoint of their influence on forest revenues. The basic scenario was based on typical forest management practices currently used in production forests in Finland [27], while the carbon sequestration scenario focused on nature management—maintaining the characteristics of the lekking site in a way that allows for the receipt of carbon offsets. To demonstrate the differences between the management scenarios, we modelled forest holdings based on stand data from different regions in Finland and ran simulations and optimisations for both management scenarios.

2.1. Forest Holdings in the Study

Our initial data were Metsään.fi-based forest stand data [28] from the five municipalities of Loppi, Pori, Kuopio, Pudasjärvi, and Sodankylä. Based on the thermal sum, Loppi and Pori are located within the Southern Finland zone (>1200 degree days (dd)), Kuopio

within the Central Finland zone (1000–1200 dd) and Pudasjärvi and Sodankylä within the Northern Finland zone (<1000 dd). Locations of the municipalities are shown in Figure 1.



Figure 1. Locations of the municipalities in the study.

From these stand data, we created two theoretical forest holdings, part of which belonged to a capercaillie lekking site in each municipality. The size of each forest holding was approximately 30 hectares, of which about 5 hectares was included in a lekking site. The development classes of the forest stands selected in the study agreed with the average values of the region in question based on data from the Natural Resources Institute Finland [29]; however, we emphasised advanced thinning stands and mature stands here because they are preferred as lekking sites by capercaillie. The percentage cover of each development class in the studied forest holdings is shown in Table 1. For the forest holdings in this study, the class of mature stands was combined with the class of advanced thinning stands due to the difficulty of identifying those two classes from stand data.

Regional average site type values were considered here to create forest holdings that were representative at the landscape-level [29]. The aim was to create one forest holding with somewhat richer site types and the other holding with somewhat drier sites than the regional average to consider the differences caused by variations in forest soil richness in each municipality. The most general tree species were Scots Pine (*Pinus sylvestris*), Norway Spruce (*Picea abies*), and Silver Birch (*Betula pendula*), depending on the site type of the forest stand. In addition, the average percentage cover of peatlands was considered, with the assumption here that the richer the site type, the less peatlands would be present within the forest holding. Site types in the forest holdings are shown in Table 2, and the comparison between the studied forest holdings and the regional average values is shown in Figure 2.

Table 1. Percentage cover of the five development classes in the forest holdings. Regional average values for each class are also shown. Abbreviation adv. is used to denote advanced.

Forest Holding	Clearcut	Young Seedling Stand	Adv. Seedling Stand	Young Thinning Stand	Adv. Thinning Stand	Mature Stand
Loppi 1	0.0%	3.0%	11.5%	26.5%	59.0%	
Loppi 2	0.0%	3.1%	11.2%	27.1%	58.6%	
Average	2.1%	5.7%	11.8%	26.0%	37.2%	17.0%
Pori 1	0.0%	5.2%	8.8%	24.2%	61.7%	
Pori 2	0.0%	3.9%	10.6%	24.2%	61.3%	
Average	1.8%	5.2%	10.6%	24.2%	41.6%	15.6%
Kuopio 1	0.0%	3.0%	10.4%	28.8%	57.8%	
Kuopio 2	0.0%	3.8%	9.6%	29.1%	57.5%	
Average	1.6%	5.6%	12.2%	29.9%	39.3%	11.4%
Pudasjärvi 1	0.0%	0.0%	11.9%	36.7%	51.4%	
Pudasjärvi 2	0.0%	1.2%	10.1%	37.4%	51.2%	
Average	1.7%	6.2%	9.4%	39.9%	32.6%	10.0%
Sodankylä 1	0.0%	0.0%	12.9%	39.5%	47.5%	
Sodankylä 2	0.0%	0.0%	12.1%	40.0%	48.0%	
Average	1.0%	7.4%	7.5%	41.1%	29.5%	11.7%

Table 2. Percentage cover of site type in the forest holdings.

	Very Rich	Rich	Damp	Sub-Dry	Dry	Barren	Sum
Loppi 1							
Mineral soils	5.5%	70.0%	7.4%	0.0%	0.0%	0.0%	82.8%
Peatlands	0.0%	1.5%	15.7%	0.0%	0.0%	0.0%	17.2%
Loppi 2							
Mineral soils	0.0%	14.3%	35.9%	22.1%	0.0%	0.0%	72.3%
Peatlands	0.0%	0.0%	17.9%	5.1%	4.5%	0.0%	27.6%
Pori 1							
Mineral soils	3.7%	45.1%	19.3%	7.4%	0.0%	0.0%	75.5%
Peatlands	0.0%	6.9%	17.6%	0.0%	0.0%	0.0%	24.5%
Pori 2							
Mineral soils	0.0%	5.0%	38.5%	19.7%	0.0%	0.0%	63.2%
Peatlands	0.0%	0.0%	0.0%	23.7%	13.1%	0.0%	36.8%
Kuopio 1							
Mineral soils	9.2%	57.4%	13.7%	0.0%	0.0%	0.0%	80.3%
Peatlands	0.0%	5.3%	10.7%	3.6%	0.0%	0.0%	19.7%
Kuopio 2							
Mineral soils	0.0%	18.8%	30.4%	17.4%	0.0%	0.0%	66.6%
Peatlands	0.0%	0.0%	21.0%	5.5%	6.8%	0.0%	33.4%
Pudasjärvi 1							
Mineral soils	0.0%	3.1%	56.4%	3.8%	0.0%	0.0%	63.3%
Peatlands	0.0%	6.0%	15.9%	14.8%	0.0%	0.0%	36.7%
Pudasjärvi 2							
Mineral soils	0.0%	0.0%	26.9%	15.1%	0.0%	0.0%	42.0%
Peatlands	0.0%	0.0%	7.2%	29.2%	21.6%	0.0%	58.0%
Sodankylä 1							
Mineral soils	0.0%	0.0%	59.6%	4.5%	3.6%	0.0%	67.6%
Peatlands	0.0%	0.0%	13.9%	18.5%	0.0%	0.0%	32.4%
Sodankylä 2							
Mineral soils	0.0%	0.0%	25.0%	31.6%	6.7%	0.0%	63.3%
Peatlands	0.0%	0.0%	2.4%	28.6%	5.8%	0.0%	36.7%

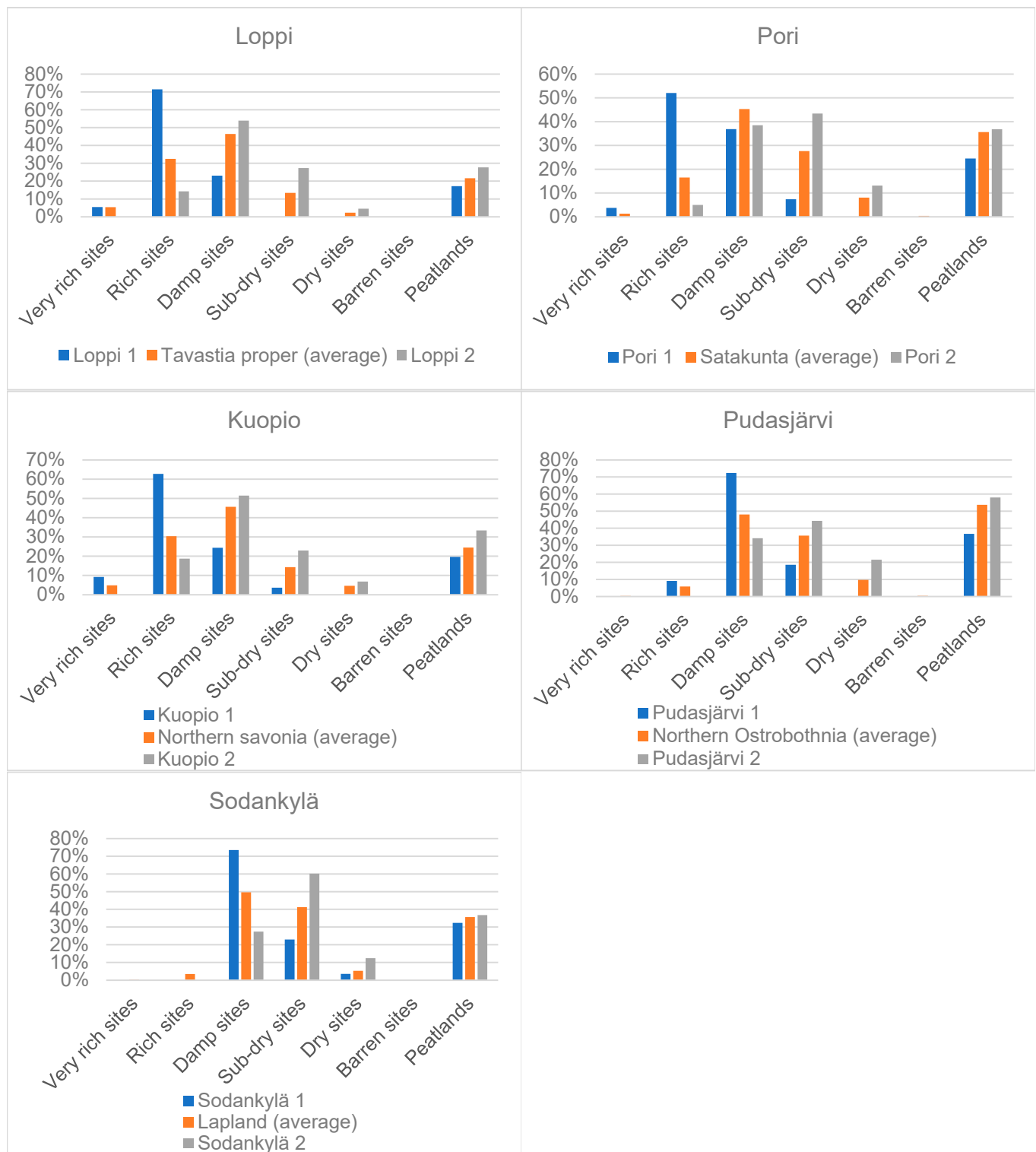


Figure 2. Percentage cover of site type and peatlands in the forest holdings in relation to the average value for the region in question. Both peatlands and mineral soils are included in the site type classes.

The stands in the lekking sites were chosen so that thinning or clear cutting, based on forest management recommendations [27], would either be in progress or would be scheduled in the next few years. Consequently, the forest owner would be in a situation where alternative management activities could be considered. Moreover, this type of situation would meet the capercaillie's requirements for lekking sites. Forest stands in the lekking sites were conifer-dominated mixed forests or pine forests. The stands outside the

lekking sites were selected as they exhibited more variation and so were considered to be more representative of real-world conditions.

2.2. Forest Planning Calculations

Forest growth and forest management were modelled using the MELA program [30], which consists of the stand simulation program MELASIM and the optimisation program MELAOPT. Simulations and optimisations were carried out using two alternative forest management scenarios, the basic scenario and the carbon sequestration scenario. The former was based on current forest management recommendations [27], and the latter was based on the maintenance of forest cover on the lekking site via delayed cuttings so that it would be possible to earn carbon offsets, a better alternative from the viewpoint of nature management. Thus, revenues were received either from timber sales alone (scenario 1) or from timber sales and the applicable carbon offsets (scenario 2). The forest planning period spans 60 years: from 2020 to 2080. To cover the range of the most general interest rates used in forest planning calculations, optimisations were carried out by maximising net present values with interest rates of 1%, 3%, and 5%. In the calculations, strategic forest management models were used.

In both scenarios, forest stands outside of the lekking sites were managed based on current forest management recommendations [27]. In Finland, forest management is typically focused on small even-aged forest stands. These stands are managed according to a regeneration cycle extending from planting or natural regeneration to the final harvesting phase. The rotation period can vary between 50 and 120 years depending on the main tree species, the thermal sum, and the site type of the stand. The main purpose is to ensure the regeneration of a productive stand of a suitable tree species for the specific site within a reasonable and cost-effective/economical time. Furthermore, younger forest stands are typically thinned out periodically from below. Uneven-aged forestry, where no final felling is performed, is seldom performed.

MELA was used to produce a finite number of feasible management schedules for each forest stand for two alternative forest management scenarios and then linear programming was applied to select stand level management schedule solutions for each stand for both scenarios. In the carbon sequestration scenario, forest management in the lekking sites was changed so that rotation period was extended by 20% and cuttings were disallowed in the first 10-year period. In addition, thinning from above was preferred as the main thinning method to extend the rotation period. The reasoning behind the carbon sequestration scenario was related to consideration of the benefits for wildlife management and biodiversity issues, as a longer rotation period enhances essential forest cover at the landscape-level [10] and increases shrub cover [31]. Simultaneously, a longer rotation period also promotes carbon sequestration [32], which makes it attractive from a climate viewpoint.

Growth models, forest management practices and timber prices were specified for each municipality by changing the MELA parameters. Stumpage prices were based on prices calculated over a ten-year period (2009–2018) to reduce the impact of annual variation, and roadside prices were determined as the sum of stumpage prices and the average cost of mechanised harvesting [30]. Forest management costs were based on average costs during the 2009–2018 period, the average earnings of the loggers, and the price of seeds and seedlings in 2018. The cost of harvesting was also based on real-time data and was the same for whole country [30]. Forest management parameters were based on forest management recommendations [27], while the growth models utilised the INKA and TINKA sample plots (for mineral soils) or the SINKA sample plots (for peatlands), which were calibrated based on National Forest Inventories [30,33]. Revenues from carbon offsets were estimated based on the modelled growth of the carbon mass in trees [29] and comparisons between the price per tonne of carbon dioxide (tCO_2) sequestered by forests in Finland. The baseline price was determined as EUR 26/ tCO_2 [34,35]. As the price of CO_2 constantly fluctuates, we also employed one lower (EUR 23/ tCO_2) and two higher prices (EUR 29/ tCO_2 and EUR 32/ tCO_2).

3. Results

3.1. Net Present Values

In comparison to the basic scenario, the carbon sequestration scenario often had a negative influence on net present values (NPV) when carbon offsets were not considered. However, at the 1% interest rate, the influence was positive in three of the ten forest holdings and in one of the holdings at the 3% interest rate. On average, NPV for the forest holdings were reduced by 0.2% at the 1% interest rate, by 2% at the 3% interest rate, and by 7% at the 5% interest rate. Rather low influences can be explained by the small area of the lekking site in relation to the forest holding, for which the net present value was calculated (Table 3).

Table 3. Net present values (NPV) and the influence of the carbon sequestration scenario in each forest holding at interest rates of 1%, 3%, and 5% (carbon offsets excluded).

		1% Interest Rate	3% Interest Rate	5% Interest Rate
Loppi 1	NPV (EUR)	1,693,364	524,008	308,358
	Influence (EUR)	+19,460	+1141	−2817
	Influence (%)	1.1%	0.2%	−0.9%
Loppi 2	NPV (EUR)	1,300,246	410,779	260,877
	Influence (EUR)	−755	−253	−6881
	Influence (%)	−0.1%	−0.1%	−2.6%
Pori 1	NPV (EUR)	1,253,893	343,217	222,254
	Influence (EUR)	−8792	−3912	−8743
	Influence (%)	−0.7%	−1.1%	−3.9%
Pori 2	NPV (EUR)	947,694	240,524	142,816
	Influence (EUR)	−3289	−1786	−5808
	Influence (%)	−0.3%	−0.7%	−4.1%
Kuopio 1	NPV (EUR)	1,147,248	320,834	194,783
	Influence (EUR)	−29,411	−10,624	−4144
	Influence (%)	−2.6%	−3.3%	−2.1%
Kuopio 2	NPV (EUR)	887,319	269,471	175,647
	Influence (EUR)	−1175	−7334	−15,562
	Influence (%)	−0.1%	−2.7%	−8.9%
Pudasjärvi 1	NPV (EUR)	427,966	120,912	75,989
	Influence (EUR)	−2061	−1338	−8219
	Influence (%)	−0.5%	−1.1%	−10.8%
Pudasjärvi 2	NPV (EUR)	301,063	83,543	53,821
	Influence (EUR)	−2345	−2038	−5982
	Influence (%)	−0.8%	−2.4%	−11.1%
Sodankylä 1	NPV (EUR)	182,470	58,690	38,349
	Influence (EUR)	+1506	−3249	−5501
	Influence (%)	0.8%	−5.5%	−14.3%
Sodankylä 2	NPV (EUR)	188,297	69,662	50,058
	Influence (EUR)	+2746	−2324	−5463
	Influence (%)	1.5%	−3.3%	−10.9%

3.2. Carbon Sequestration

As expected, the carbon sequestration scenario resulted in a greater carbon mass in the trees, at least at the beginning of the 60-year period. In the later years, the influence will fluctuate depending on timing of the cuttings—the later 10-year periods were emphasised due to the investment in carbon sequestration rather than the beginning of the 60-year period. This was projected to sometimes result in a decrease in carbon mass, especially after the year 2050. In addition to cuttings, natural disturbances and climatic changes could increase the fluctuation in carbon mass, particularly in the long term. The increment in total carbon mass in trees caused by the carbon sequestration scenario is shown in Table 4.

Table 4. The difference between the carbon sequestration scenario and the basic scenario in total carbon mass stored in the trees (in tonnes) at interest rates of 1%, 3%, and 5%.

Forest Holding	Interest Rate	2030	2040	Year 2050	2060	2070	2080
Loppi 1	1%	140.08	252.99	343.07	−86.36	−359.46	−323.96
	3%	207.26	496.04	−67.96	−42.33	−108.26	−69.40
	5%	264.98	13.79	−11.90	−34.33	−17.72	−12.85
Loppi 2	1%	106.60	175.51	345.33	−45.19	−94.93	−613.97
	3%	242.68	2.36	−30.03	−45.00	−29.75	−14.97
	5%	412.97	−4.67	−24.09	−39.43	−10.32	−46.50
Pori 1	1%	184.59	221.34	106.33	−82.59	−578.25	−864.04
	3%	230.76	−27.90	30.47	27.23	9.68	−20.72
	5%	416.44	33.63	10.34	46.55	−95.54	−81.27
Pori 2	1%	70.16	123.18	29.27	21.46	84.41	91.57
	3%	155.72	187.80	−172.10	88.46	10.05	20.37
	5%	187.03	291.67	70.20	40.12	−102.52	−99.17
Kuopio 1	1%	117.72	245.95	259.59	224.76	56.23	−716.66
	3%	150.48	235.28	−0.48	−364.55	11.96	44.89
	5%	202.47	−204.49	−2.71	11.81	45.53	93.45
Kuopio 2	1%	181.00	238.62	118.73	102.10	−153.49	−385.77
	3%	63.41	93.26	1.72	−126.50	−32.91	−37.88
	5%	310.62	276.92	6.90	267.40	−75.55	−134.16
Pudasjärvi 1	1%	105.03	110.62	113.40	113.53	197.81	199.49
	3%	155.89	133.16	159.99	−0.08	−5.25	−18.74
	5%	174.84	365.13	375.00	399.62	−74.45	−140.27
Pudasjärvi 2	1%	57.26	54.39	48.62	42.16	36.93	33.22
	3%	142.29	138.46	86.64	−51.17	−68.76	−91.67
	5%	189.23	233.85	130.17	138.47	105.10	72.48
Sodankylä 1	1%	0.00	51.49	−17.12	−1.45	11.05	14.74
	3%	121.45	60.94	150.20	155.77	157.54	−64.48
	5%	177.83	105.26	155.34	171.94	200.22	−1.18
Sodankylä 2	1%	14.83	36.58	−101.96	−38.79	−42.42	−26.03
	3%	263.66	94.32	−13.95	−33.25	−52.05	−55.04
	5%	327.53	93.68	−16.56	−37.19	−55.83	−58.97

3.3. Carbon Offsets

When the price of CO₂ was EUR 23 or EUR 26/tonne, the carbon sequestration scenario had a negative influence on NPV during the first 10-year period, with the exception of the forest holdings in Pudasjärvi and Sodankylä (Table 5). At higher CO₂ prices, the influence was also positive in some more southern forest holdings. In Sodankylä 1, the carbon sequestration scenario did not change forest management (compared to the basic scenario) in the first 10-year period (if optimised at the 1% interest rate) and resulted in no influence on net revenues. The increase in the interest rate led to either a lower or higher influence on net revenues, depending on how this changed forest management in the logging site. (Table 5).

However, when the influence on average revenues was considered in the subsequent six 10-year periods, the values were mainly positive. The decrease in net revenues was only apparent in a very few cases, regardless of the price of CO₂ or the interest rate. In a few forest holdings, lower carbon sequestration prices might result in greater net revenues over the 60-year period due to more intense cutting at the end of the rotation period and a loss of carbon mass as a consequence. This was evident in four holdings when the 1% interest rate was used and in one holding when the 3% interest rate was used. Again, changes in the interest rate could lead to notably different types of forest management, which can be seen as variation in the influence of the interest rate (Table 6).

Table 5. The influence of the carbon sequestration scenario on net revenues (in euros) during the first 10-year period at carbon prices of EUR 23/tCO₂, EUR 26/tCO₂, EUR 29/tCO₂ and EUR 32/tCO₂ (carbon offsets included).

	Interest Rate	EUR 23/tCO ₂	EUR 26/tCO ₂	EUR 29/tCO ₂	EUR 32/tCO ₂
Loppi 1	1%	−5368.4	−3811.9	−2255.5	−699.1
	3%	−8186.5	−5883.4	−3580.7	−1277.9
	5%	−12,664.7	−9720.4	−6776.2	−3832.0
Loppi 2	1%	−1764.2	−579.7	604.8	1789.2
	3%	−14,182.3	−11,485.9	−8789.4	−6092.9
	5%	−20,623.3	−16,034.7	−11,446.2	−6857.7
Pori 1	1%	−10,674.4	−8623.4	−6572.3	−4521.3
	3%	−11,566.7	−9002.6	−6438.7	−3874.7
	5%	−20,969.8	−16,342.7	−11,715.7	−7088.6
Pori 2	1%	−2531.7	−1752.2	−972.7	−193.2
	3%	−4824.0	−3094.2	−1363.5	366.7
	5%	−4719.8	−2641.7	−563.5	1514.6
Kuopio 1	1%	−4122.1	−2814.1	−1506.1	−198.1
	3%	−5322.3	−3650.4	−1978.3	−306.3
	5%	−8235.7	−5986.1	−3736.4	−1486.8
Kuopio 2	1%	−8056.8	−6045.8	−4034.7	−2023.6
	3%	−27,569.4	−26,864.9	−26,160.3	−25,455.7
	5%	−14,085.7	−10,634.3	−7183.0	−3731.6
Pudasjärvi 1	1%	1145.0	2312.0	3479.0	4646.0
	3%	2903.5	4636.1	6367.7	8099.9
	5%	2964.8	4907.4	6850.1	8792.8
Pudasjärvi 2	1%	1721.9	2358.1	2994.4	3630.6
	3%	7916.0	9497.2	11,078.0	12,659.0
	5%	11,014.7	13,117.2	15,219.8	17,322.4
Sodankylä 1	1%	0.0	0.0	0.0	0.0
	3%	1227.7	2577.4	3926.6	5276.1
	5%	1173.2	3149.0	5124.9	7100.7
Sodankylä 2	1%	2371.2	2536.0	2700.8	2865.5
	3%	1120.9	4050.0	6980.0	9909.6
	5%	2243.7	5882.9	9522.2	13,161.4

Table 6. Influence of the carbon sequestration scenario on average net revenues (in euros) during the six 10-year periods at carbon prices of EUR 23/tCO₂, EUR 26/tCO₂, EUR 29/tCO₂ and EUR 32/tCO₂ (carbon offsets included).

	Interest Rate	EUR 23/tCO ₂	EUR 26/tCO ₂	EUR 29/tCO ₂	EUR 32/tCO ₂
Loppi 1	1%	13,315.2	13,252.9	13,190.6	13,128.3
	3%	12,379.3	13,148.4	13,917.6	14,686.8
	5%	4716.0	5090.1	5464.1	5838.1
Loppi 2	1%	23,451.2	23,216.6	22,982.1	22,747.6
	3%	2972.8	3204.8	3436.8	3668.8
	5%	9514.9	10,048.1	10,581.4	11,114.7
Pori 1	1%	15,094.5	13,219.3	11,344.1	9468.8
	3%	3547.7	4009.7	4471.8	4933.9
	5%	8810.6	9422.0	10,033.3	10,644.7
Pori 2	1%	4365.3	5143.2	5921.0	6698.9
	3%	4114.1	4651.6	5189.2	5726.8
	5%	10,258.8	10,976.1	11,693.4	12,410.6
Kuopio 1	1%	25,115.1	25,462.5	25,809.9	26,157.2
	3%	−4401.7	−4258.0	−4114.4	−3970.7
	5%	451.5	722.0	992.5	1263.0

Table 6. *Cont.*

	Interest Rate	EUR 23/tCO ₂	EUR 26/tCO ₂	EUR 29/tCO ₂	EUR 32/tCO ₂
Kuopio 2	1%	15,441.9	15,629.2	15,816.6	16,004.0
	3%	3127.9	3055.8	2983.8	2911.8
	5%	22,014.5	23,851.8	25,689.0	27,526.3
Pudasjärvi 1	1%	8434.4	9989.7	11,545.1	13,100.4
	3%	7409.5	8196.5	8983.5	9770.5
	5%	21,862.4	23,899.2	25,936.0	27,972.7
Pudasjärvi 2	1%	3344.1	3848.9	4353.7	4858.5
	3%	−365.0	−76.5	212.0	500.6
	5%	13,576.5	15,186.3	16,796.1	18,405.9
Sodankylä 1	1%	422.6	531.3	640.1	748.8
	3%	10,965.3	12,042.0	13,118.7	14,195.4
	5%	14,631.0	16,129.9	17,628.8	19,127.6
Sodankylä 2	1%	−2275.4	−2567.6	−2859.8	−3152.0
	3%	4226.5	4603.7	4980.9	5358.1
	5%	5346.0	5813.9	6281.8	6749.7

4. Discussion

To find new solutions for the cost-effective natural management of production forests, we demonstrated the potential of carbon offsets to compensate for the possible losses in forest revenues caused by the management of capercaillie lekking sites. Our study was based on simulations and optimisations for two alternative forest management scenarios in ten forest holdings located in five different regions of Finland. The basic scenario was based on typical forest management that followed forest management recommendations [27], and the carbon sequestration scenario aimed to enhance biodiversity maintenance and earn payments for carbon sequestration.

In comparison to the basic scenario, the influence of the carbon sequestration scenario on NPV was generally negative when carbon offsets were not considered. This would indicate that the maintenance of lekking sites typically caused losses in revenue from timber sales. Exceptions appeared only in very few cases. For most of the forest holdings, the loss of income increased when the interest rate increased, although in Kuopio 1, the loss was substantially less at the 5% interest rate than at the lower interest rates. Even if the decrease in net revenues might be rather low compared to the NPV of the forest holding, economic losses could be considerable from the viewpoint of forest owner. In our calculations, forest management only changed for 5 hectares of the total forest holding of 30 hectares.

As expected, the carbon sequestration scenario typically resulted in a greater storage of carbon in the trees than in the basic scenario, at least in the short-term, because of delayed cuttings. Carbon offsets based on the increment in total carbon mass in the trees completely compensated for losses in income within the first 10-year period in the forest holdings in Pudasjärvi and Sodankylä, regardless of the price of CO₂ (between EUR 23 and EUR 32/tonne). In the other municipalities, the influence during the first ten years was mainly negative except for some cases with higher CO₂ prices. However, when the full 60-year period was considered, the carbon sequestration scenario was an economically better choice in every forest holding at two of the three alternative interest rates. Furthermore, for most of the forest holdings, a negative influence was not observed at any of the interest rates used. Since a change in the interest rate could lead to notably different forest management, the increased interest rates did not appear to be similar in different forest holdings. However, carbon offsets were, in general, greater than the losses recorded from timber sales.

As income is one of the most important aims of forest use for forest owners [36,37], our results can be seen as encouraging. According to Ahtikoski et al. [38], the price of carbon also encourages the use of continuous-cover forestry with less intense harvesting, which is in line with our results. The CO₂ prices used in this study were rather cautious [39]; consequently, our results should not overestimate the income from carbon sequestration. On the other hand, the use of higher prices in the calculations could have made the

carbon sequestration scenario an even better alternative than our results have indicated. In addition, future pricing of carbon sequestration will more likely rise than decline [40,41]. For some forest owners, even lower prices of CO₂ may be sufficient to make the carbon sequestration scenario preferable in their decision making, as they value other aims, such as the good condition of habitats, recreation potential of forests, or climate benefits [37,42,43].

The forest holdings that we studied are located across a wide range of typical forest growth conditions in Finland. Our results suggest that the carbon sequestration scenario might be economically reasonable regardless of the thermal sum (i.e., geographic location) or forest site type. However, a remarkable change in climatic conditions could alter the growth of trees, which could change our results. In addition, it must be noted that our calculations were limited to only considering the carbon in trees, even though a significant amount of carbon is stored in the forest soil and in harvested wood products and those also play crucial roles in the carbon cycle [44]. To comprehensively assess the carbon compensation, all carbon pools (biomass, soil, litter, harvested products) should be considered. Peatlands play a critical role in climate issues because of the enormous amount of soil carbon stored within them, and so forest management activities have a considerable influence on carbon emissions from these soils [45]. Naturally, differences in soil carbon content between peatlands and mineral soils cannot be seen in our calculations, which focused solely on the carbon mass in the trees. Furthermore, variations in the prices of timber and carbon sequestration could temporarily change the situation.

In addition to capercaillie lekking sites, carbon offsets could be cost-effectively utilised in nature management to maintain the habitats of a wide range of species that require adequate forest cover. Moreover, Schuster et al. [46], Matzek et al. [47], and Wolfe and Elizondo [48] have suggested that carbon offsets could notably reduce conservation costs, which is in line with our results. These studies also indicate that our results might apply not only to Finland, but also to other regions. From the viewpoint of grouse species, the maintenance of forest cover could be beneficial in many ways; for example, it could decrease the need for ditch cleaning [49], which in turn, could decrease chick mortality rates [50,51]. In addition, the avoidance of clear cutting could enhance the growth of blueberry shrubs [31], which are vital for grouse species [52–55]. In addition, the enhanced growth of blueberry could benefit many other species [56], increase the recreational value of the forests, and have notable economic value [57].

5. Conclusions

Even though the maintenance of lekking sites can result in decreased revenue from timber sales, the decrease is often so small that the receipt of carbon offsets could offer an economically better option, particularly in the long term. Furthermore, the possibility of earning additional income from carbon sequestration does not seem to be restricted to any specific geographic location. The price of carbon may rise in the future, which could make it an even more attractive choice for forest owners, encouraging positive attitudes towards biodiversity conservation, the recreation potential of forests, and climate change mitigation. We introduced an approach to the multiple uses of forests, which could be utilised in future decision making in the forestry sector. Our results could also provide worthwhile topics for future research. We recommend that forest carbon stores should be considered more comprehensively in future carbon offset calculations. Our results are encouraging, not only from the viewpoint of forest revenues but also for the multiple uses of forests and nature management in general.

Author Contributions: Conceptualization, A.H., J.M., P.I. and A.P.; methodology, A.T., A.H. and J.M.; software, A.T. and A.H.; validation, A.T.; formal analysis, A.T.; investigation, A.T.; resources, A.H.; writing—original draft preparation, A.T.; writing—review and editing, A.H., J.M., P.I. and A.P.; visualization, A.T., J.M. and P.I.; supervision, A.P.; project administration, P.I.; funding acquisition, A.P. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by Ministry of Agriculture and Forestry in Finland R&I Programme “Catch the Carbon” under project “Carbon neutral use of cut-away peatlands” and the Academy of Finland Flagship “Forest-Human-Machine Interplay—Building Resilience, Redefining Value Networks and Enabling Meaningful Experiences” (UNITE) under Grant (337127) and (337655).

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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